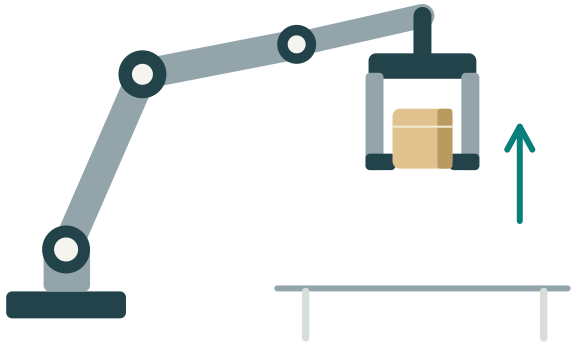
When expected physics meets the real world

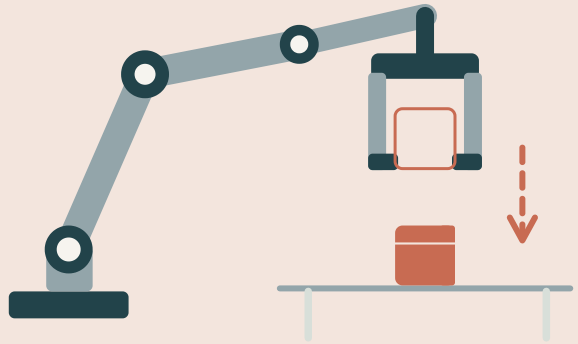
How can robots handle deployment conditions they did not encounter during training?

Expected conditions



Mismatch

Actual conditions



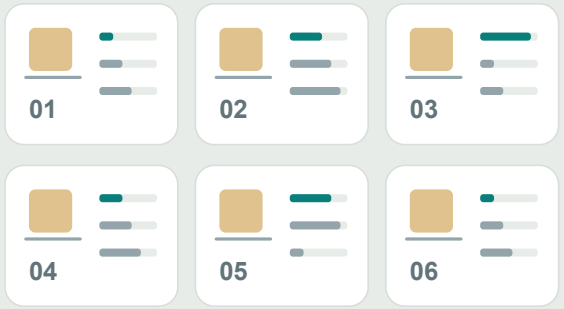
Physical conditions can differ from those assumed during training.

Use an estimate of the actual conditions

A policy trained across varied physical conditions could execute successfully when supplied with a sufficiently accurate estimate of its environment.

TRAINING

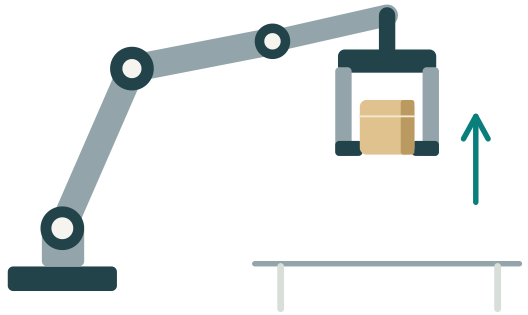
Varied physical conditions



Trained policy

Physical estimate

Fixed pick-and-lift task



AT DEPLOYMENT

Estimate the conditions the robot actually faces.

What we mean by understanding

Estimating task-relevant physical properties well enough to support task execution.

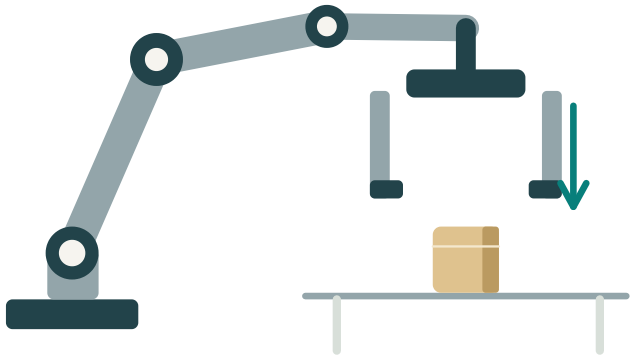
Objective	Grasp strategy	Physical understanding
Pick and lift	**Top-down grasp**	**Initial estimate**
FIXED	FIXED	STUDIED

Fixing the objective and strategy keeps the focus on physical estimation.

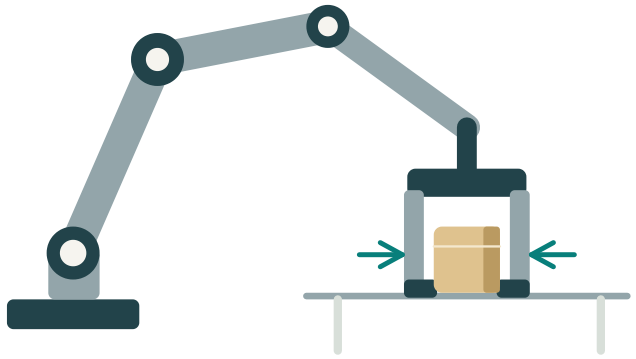
One task. One grasp strategy.

Use a top-down pick and lift to isolate the physical-estimation question.

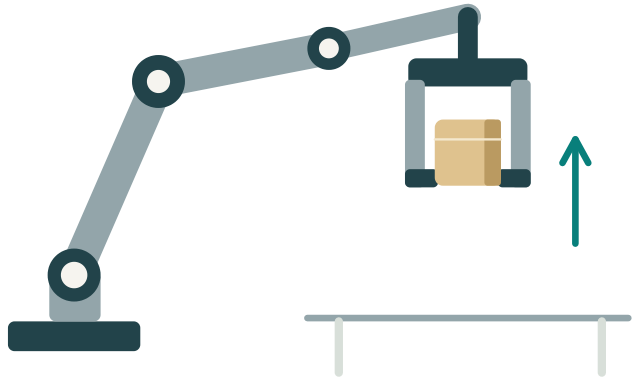
1 Approach



2 Grasp



3 Lift



The objective and grasp strategy remain fixed throughout the experiment.

Prepare the policy in simulation

Select a compact physical context, then train across variations in that context.

1 Select relevant variables

mass

friction

geometry

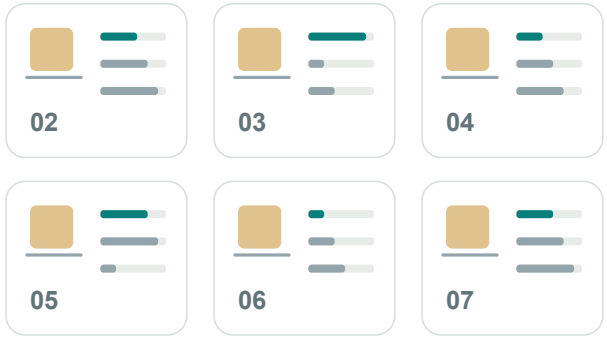
…

Illustrative candidate properties

CREST

Keep the representation and computation manageable.

2 Vary physical conditions



Sample varied combinations in the simulator.

3 Train a conditioned policy

State + known physical context



Policy training

Learn execution across the modeled conditions.

Selected variables provide an approximation of the physical environment.

Method reference: CREST · Lee et al., ICRA 2021 · arxiv.org/abs/2103.16772

Form the initial estimate at deployment

Study how initial observations become a useful approximation of physical conditions.

Observe



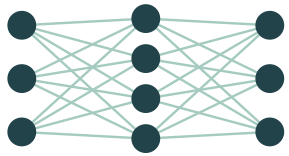
Initial observations

Estimate



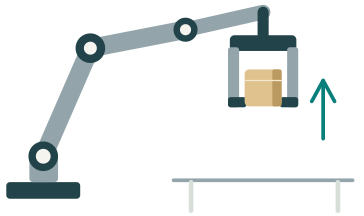
Selected physical properties

Condition



Policy trained in simulation

Execute



Fixed pick and lift

CURRENT FOCUS

How is the initial estimate formed?

Test whether sufficiently accurate estimates support successful execution in simulation.

Initial understanding comes first

Start with the initial estimate; investigate revision through interaction in a later stage.

CURRENT EXPERIMENT

Form an initial physical estimate.

Initial estimate → Task execution

LATER EXTENSION

Use observed responses to revise the estimate.

Success, failure, and interaction data

Revised estimate --> Next execution

Can a compact physical estimate support reliable execution of a fixed manipulation task?